

PX4Flow VELOCITY VALIDATION

Final Internship Report and Handover Summary

Project: PX4Flow -> i.MX7 Optical-Flow Logging -> Cube/Pixhawk Validation

Organization	General Aeronautics Pvt Ltd
Project Area	GPS-denied velocity estimation feasibility
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1. Executive Summary

This project evaluated whether a PX4Flow optical-flow sensor can provide useful horizontal velocity information for GPS-denied operation and possible future use on the Krishak 2i platform. The work covered hardware integration, independent data logging, time synchronization, automated file transfer, offline signal processing, controlled validation tests, and preparation of a repeatable test workflow.

Operating-range bike tests were conducted at nominal speeds of 15 and 20 km/h, representative of the approximately 4–5 m/s speed range used during spraying. Across four verified steady-motion windows, the optical-flow median-speed errors were 1.5%, 15.4%, 15.8% and 11.2%. All four remained within the project's 20% feasibility criterion, with an average absolute error of approximately 11.0%. Additional long-duration tests at a nominal 17 km/h and deliberate pitch, roll and yaw disturbances produced mixed repeatability, especially during roll and yaw. The project therefore demonstrates useful operating-range feasibility under selected clean conditions while identifying rigid mounting, stable height measurement and vibration/environment testing as the next requirements before integration approval.

A major final improvement was making the i.MX7 independent of the laptop during field logging. It now connects to a phone hotspot, synchronizes time, starts the optical-flow logger automatically, indicates status through the onboard LED, and supports passwordless log download through the laptop analysis menu.

2. Problem Statement and Objectives

Agricultural drones may experience degraded GNSS performance at low altitude or near structures. Optical flow can provide short-range horizontal motion information when GPS is weak or unavailable. The purpose of this internship was to determine whether the available PX4Flow hardware can produce sufficiently stable and useful velocity estimates for further GPS-denied development.

2.1 Main objectives

- Interface PX4Flow with the TechNexion PICO-PI i.MX7 through UART.
- Log optical-flow data independently into CSV files on the i.MX7.
- Collect Cube/Pixhawk BIN logs containing GPS, IMU and EKF data.
- Process optical-flow data using scale, height and gyro compensation.
- Validate velocity using measured distance divided by video duration.
- Use GPS/EKF only as secondary trend references when their data are valid.
- Create a repeatable field workflow that does not require carrying a laptop.

2.2 Optical Flow Sensor Selection

The sensor selected for this project was the **PX4Flow v1.3.1 optical-flow smart camera**. It is a downward-facing vision sensor designed to estimate horizontal motion relative to the ground by tracking the displacement of visible surface features between successive image frames.

Unlike a conventional camera that sends complete images to another processor, PX4Flow performs the optical-flow calculation directly on the sensor board. It uses a **752 × 480 pixel global-shutter image sensor**, an onboard **168 MHz ARM Cortex-M4F processor** and an integrated three-axis gyroscope. The gyroscope measurements support compensation for apparent image motion caused by pitch, roll and yaw rotation. Processed optical-flow and gyro information can be transmitted through **UART or I2C** interfaces.

PX4Flow measures angular image displacement rather than ground velocity directly. Therefore, the measured optical-flow rate must be combined with the sensor height above the ground and corrected for rotational motion to obtain horizontal velocity:

$$V_{OF} = h \cdot \sqrt{\{(\theta_{x,flow} - \omega_x)^2 + (\theta_{y,flow} - \omega_y)^2\}}$$

where:

- V_{OF} is the estimated horizontal speed,
- h is the sensor height above the surface,
- $\theta_{x,flow}$ represents the measured angular optical-flow rate, and
- ω represents the corresponding gyro rotational rate.

2.2.1 Optical Flow Sensor Selection

PX4Flow v1.3.1 was selected as the evaluation sensor for the following reasons:

Selection factor	Relevance to the project
Onboard optical-flow processing	Reduced the image-processing load on the i.MX7 and allowed the project to work directly with processed flow measurements.
Integrated three-axis gyroscope	Enabled investigation of gyro compensation during pitch, roll and yaw disturbances.
UART/MAVLink output	Allowed direct connection to the PICO-PI i.MX7 and independent CSV logging without initially integrating the sensor into the Cube flight controller.
Global-shutter camera	Better suited to motion measurement than a rolling-shutter camera because the complete image frame is captured at the same instant.
High-rate measurements	Provided sufficient temporal resolution for stationary, hand-carried and bike-speed motion experiments.
Low-light sensitivity	Made the sensor usable across a wider range of indoor and outdoor lighting conditions without requiring an illumination LED.
Pixhawk and autopilot ecosystem support	Existing PX4 and ArduPilot documentation provided a known path for future flight-controller integration and GNSS-denied navigation experiments.

Selection factor

Relevance to the project

Available and reprogrammable hardware The sensor was already available for evaluation, and its open firmware and documented interfaces simplified prototyping and troubleshooting.

PX4Flow is specifically intended for downward-looking ground-motion estimation and has documented use in non-GNSS navigation systems. This made it a suitable platform for evaluating low-altitude horizontal velocity estimation for possible future application on the Krishak 2i platform.

The sensor was selected as a practical feasibility platform, not as the result of a complete commercial comparison of all available optical-flow sensors. Consequently, this project evaluates whether the existing PX4Flow hardware can provide sufficiently accurate and repeatable velocity measurements; it does not conclude that PX4Flow is the optimum sensor for final production integration.

3. System Architecture

Subsystem	Function
PX4Flow	Downward-facing smart camera that calculates optical flow onboard, measures rotational motion using its integrated gyroscope, and outputs flow and gyro data through MAVLink over UART
i.MX7	Reads PX4Flow UART data, creates CSV logs, manages hotspot connection and time synchronization.
Cube/Pixhawk	Creates BIN logs containing GPS, IMU, EKF and vehicle timing data.
Phone hotspot	Provides network access for i.MX7 time synchronization and laptop-to-i.MX7 SSH transfer.
Laptop post-processing tool	Stops the logger, downloads new CSV files, matches logs, prints validation results and creates plots.
Video + measured distance	Provides the primary reference speed for controlled straight-line tests.

Final field workflow: Phone hotspot ON -> power the setup -> i.MX7 connects and synchronizes time -> inner LED becomes solid -> perform the test -> connect laptop to the same hotspot -> stop logger and download only new CSV files -> run validation and plots.

4. Major Work Completed

4.1 Hardware and basic logging

- Corrected the UART wiring between PX4Flow and i.MX7 and verified live MAVLink data.
- Developed and tested `/root/of_reader.py` for readable optical-flow logging.
- Created numbered CSV files in `/root/of_logs` using the format `Test_<number>_<date>_<time>.csv`.
- Integrated the i.MX7, PX4Flow, Cube/Pixhawk and GPS on a common test frame.

4.2 Independent field logging and automation

- Changed the i.MX7 from access-point mode to Wi-Fi client mode and saved the phone hotspot connection.
- Disabled the old service that forced the i.MX7 back into access-point mode after reboot.

- Used ConnMan NTP adjustment as the gate before starting the logger.
- Configured the inner onboard LED: fast blink while waiting, solid while logging, slow blink on error, OFF after intentional stop.
- Configured passwordless ed25519 SSH access from the Windows laptop.
- Added dynamic i.MX7 IP discovery by scanning the active hotspot subnet.
- Automated safe logger stop, filesystem sync, new-file detection and selective SCP download.

4.3 Post-processing and plotting

- Separated motion-window and zero-velocity validation into dedicated CSV configuration files.
- Added support for multiple motion windows from the same optical-flow CSV file.
- Calculated the official OF median using all full-rate OF samples inside the selected motion window.
- Removed misleading full-log and auto-zoom plots from the main workflow.
- Generated one dedicated native-rate plot per configured window containing every valid OF, GPS and EKF sample, the analysis boundaries, reference-speed line and sensor median lines.
- Stored explicit GPS and EKF time offsets in the window CSV files and applied them before window selection, statistics and plotting. A simplified GPS-based automatic full-run window was also added for long-duration robustness tests.

5. Validation Method

5.1 Primary motion reference

For controlled straight-line motion, the reference speed is calculated from the measured distance and the edited video duration. The same video is used to identify the true crossing interval. This prevents stationary, acceleration and deceleration portions from corrupting the official median.

Reference speed = measured distance / video duration

5.2 Zero-velocity and rotation-only tests

For stationary, pitch-only, roll-only and yaw-only tests, expected translational velocity is zero. Percentage error is therefore not used. The correct outputs are median, mean and maximum apparent OF speed, plus the sample count.

5.3 Secondary references

GPS and EKF speed are used as secondary trend or sanity references because their rates, filtering and timing do not perfectly match the independently logged i.MX7 optical-flow data. For the bike experiments, the maintained speed shown on the bike speed indication was used as a nominal operating-point reference. These tests assess feasibility and repeatability at 15-20 km/h; they are not treated as laboratory-grade absolute calibration.

6. Key Results

6.1 Latest valid straight-line results

Test	Reference speed	OF median	Absolute error	Error
Slow test 3	0.806 m/s	0.742 m/s	0.064 m/s	8.0%
Medium test 3	1.382 m/s	1.262 m/s	0.120 m/s	8.7%
Fast test 3	2.419 m/s	2.197 m/s	0.222 m/s	9.2%

Interpretation: The latest valid slow, medium and fast windows are consistent and remain below approximately 10% error. Earlier extreme errors were mainly caused by incorrect motion-window selection rather than a consistent failure of the optical-flow estimator.

6.2 Stationary and rotational checks

Test	Median apparent speed	Mean	Maximum	Interpretation
Stationary, untouched	0.025 m/s	0.025 m/s	0.030 m/s	Low noise floor
Pitch-only	0.021 m/s	0.186 m/s	1.561 m/s	Low median; short peaks during rotation
Roll-only	0.028 m/s	0.140 m/s	1.703 m/s	Low median; short peaks during rotation
Yaw-only	0.333 m/s	0.350 m/s	1.212 m/s	Higher false velocity; needs investigation

Interpretation: Stationary, pitch and roll median values are low. Yaw produces a noticeably higher apparent velocity and remains an open calibration or compensation issue.

6.3 Operating-Range Bike Tests and Long-Duration Robustness

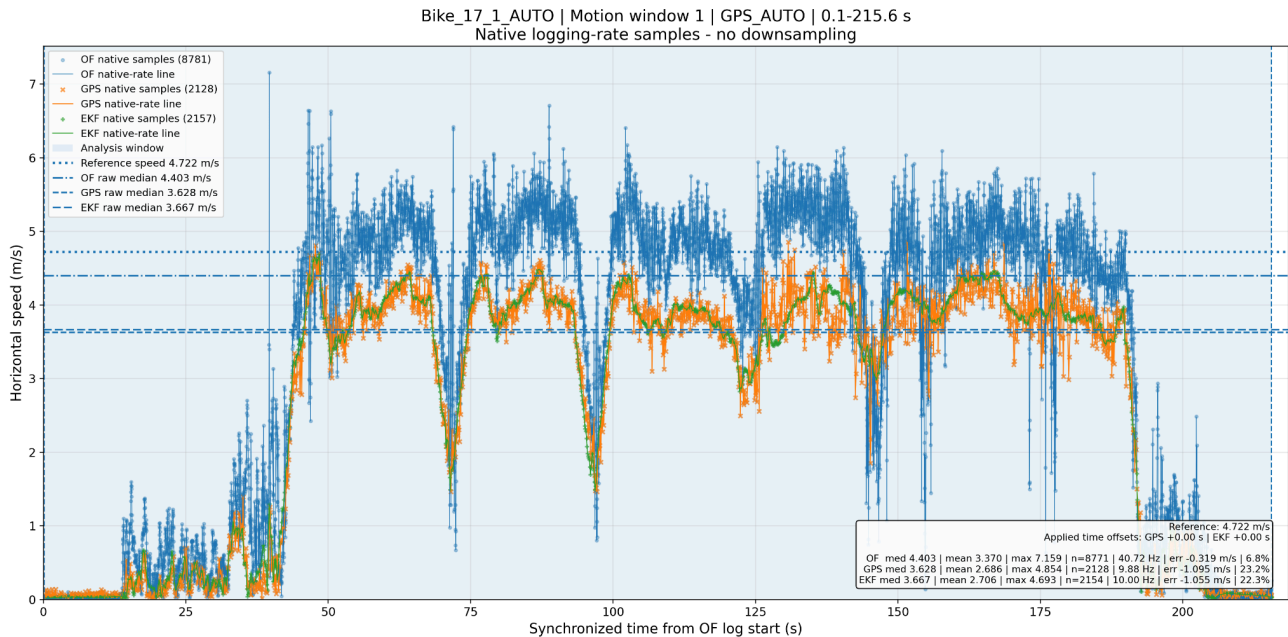
Four dynamic bike tests were conducted at nominal speeds of 15 and 20 km/h using two repeats at each speed. The PX4Flow, battery and frame were manually carried while a second person maintained the bike speed. Verified steady-motion windows were selected from the synchronized native-rate OF, GPS and EKF plots.

Across the four verified operating-range windows, the OF median-speed errors were 1.5%, 15.4%, 15.8% and 11.2%. All four met the defined 20% feasibility criterion, giving an average absolute error of approximately 11.0%. Because the speed was maintained from the bike speed indication rather than derived from an independently measured route, these values are reported as nominal operating-point results rather than laboratory-grade absolute accuracy.

A second set of five nominal 17 km/h clean runs and separate pitch, roll and yaw disturbance runs was processed using automatic first-to-last GPS stationary-state windows. GPS and EKF median speeds remained generally consistent at approximately 3.5–3.8 m/s, while PX4Flow varied between runs. Several clean and pitch-disturbance runs remained reasonably close to GPS, whereas one clean run and the roll and yaw disturbance runs showed substantial underestimation.

Overall, the verified 15–20 km/h windows demonstrate useful operating-range potential, while the longer disturbed runs show that repeatability is sensitive to orientation, height variation and hand-induced vibration. The earlier measured-distance, video-timed straight-line tests remain the controlled calibration baseline, with approximately 8–9% error across slow, medium and fast hand-carried conditions.

6.4 Latest Plots



The above plot is a velocity plot of PX4FLOW optical flow sensor. Here3+ GPS module and Extended Kalman Filter (EKF). The speedometer reading of the bike (indicated by a blue dotted line) was kept at 17 kmph (4.722 m/s). The blue graph is the optical flow sensor estimated velocity, while the orange and green graphs are the GPS and EKF estimated velocities respectively. From this we can tell that the optical flow sensor is more sensitive to any kind of vibrations caused during handling the drone frame. The blue shaded part which covers the whole plot is the selected window. I have achieved accurate syncing of both the GPS and Optical flow logs.

7. Main Problems and Corrections

Problem	Effect	Correction
Wrong or full analysis windows	Stationary and transition data changed the OF median and produced misleading errors.	Use only video-confirmed crossing intervals for official results.
Uncontrolled logger restarts	Extra, tiny or duplicate CSV files were generated.	Control logging through systemd and stop the service before downloading.
i.MX7 clock not synchronized	Wrong-date filenames such as 2022 or 2030 appeared.	Wait for ConnMan NTP adjustment before starting the logger.
Phone hotspot IP changed	The laptop could not consistently locate the i.MX7.	Scan the active /24 subnet and authenticate the correct SSH device.
Laptop required for every test	Field setup was inconvenient and slow.	Autoconnect, autologging, LED status and passwordless download were implemented.
GPS/OF graph mismatch	Curves did not perfectly overlap despite reasonable speed magnitude.	Use distance/video as the primary reference and GPS only as a secondary trend.

8. Current Status and Limitations

8.1 Current status

- PX4Flow-to-i.MX7 logging is operational and field-ready for further testing.
- Time synchronization, automatic logger start, LED status and passwordless download are working.
- The analysis tool supports multiple windows and full-rate OF median calculation.
- Four verified nominal 15–20 km/h bike windows met the defined 20% feasibility criterion, with an average absolute error of approximately 11.0%.
- Automatic GPS-bounded long-duration processing, historical-date reproduction and final native-rate plotting were completed for handover.

8.2 Remaining limitations

- The nominal bike-speed reference was not supported by an independently measured route, so it remains an operating-point estimate rather than laboratory ground truth.
- Yaw-only false velocity is higher than pitch and roll.
- Performance under actual drone vibration and mounting conditions has not been validated.
- Surface texture, lighting, height variation and weather sensitivity require additional testing.
- The current work supports feasibility, not final Krishak 2i integration approval.

9. Current Work

Day	Work Done	Output of the Day
Day 36 - 20 July	Tested autonomous i.MX7 logging using the mobile-hotspot method. The i.MX7 searched for the configured hotspot, synchronized its clock and automatically started Optical Flow logging. This reduced the Cube-i.MX7 logging delay from approximately 1–2 minutes to around 11–12 seconds. Stationary, pitch, roll and yaw tests were also performed to evaluate false velocity during rotational motion.	Hotspot-based time synchronization and automatic logging worked successfully. Stationary and low-disturbance tests produced false velocities in the cm/s range. Pitch, roll and yaw introduced larger transient apparent velocities, confirming the need for correct gyro compensation, stable height and sensor orientation. The setup was ready for dynamic bike testing
Day 37 - 21 July	Conducted four dynamic bike tests: two at 15 km/h and two at 20 km/h. The PX4Flow, battery and frame were manually carried while the bike speed was maintained by a second person. The tests included height variation, hand vibration and pitch, roll and yaw disturbances. Native-rate OF, GPS and EKF data were synchronized and analysed without downsampling. Verified motion windows were selected for each run.	All four verified windows met the defined 20% acceptance criterion. OF median-speed errors were 1.5%, 15.4%, 15.8% and 11.2% , with an average absolute error of approximately 11.0% . One test was classified GOOD and three were ACCEPTABLE .
Day 38 - 22 July	Cleaned and consolidated the PX4Flow post-processing codebase. Removed obsolete cache files and identified legacy scripts for archival. Added a user-selectable DDMMYYYY test-date filter so historical logs can be processed without depending on the current date. Corrected Cube/OF log collection and matching issues, including the Cube BIN loop and active time_match.py file. Added explicit GPS and EKF time-offset columns to the motion-window CSV. Simplified the unified plotter to generate only native logging-rate OF, GPS and EKF graphs, with no smooth-graph output or hidden downsampling. Updated the README to document the current workflow, file structure, date filtering, alignment settings and testing procedure.	The final workflow is now cleaner and more reproducible. Historical tests can be selected by date, synchronization offsets are explicitly stored per test window, and new tests safely default to zero time shift. The plotter retains every valid native-rate sample and produces only the required full-rate graphs. The cleaned code and documentation are ready for final end-to-end testing and handover preparation.

Day	Work Done	Output of the Day
Day 39 - 23 July	Implemented and tested the simplified GPS-timestamp automatic window-selection method using the first and last stationary GPS states. Processed eight matched Cube/OF pairs from five nominal 17 km/h runs and separate pitch, roll and yaw disturbance tests using native-rate OF, GPS and EKF data.	Automatic matching, full-run window selection, statistics and plotting completed successfully. GPS and EKF remained consistent, while OF repeatability varied; clean and pitch runs were closer, whereas roll and yaw showed larger underestimation. The results were retained as robustness and repeatability evidence.
Day 40 - 24 July	Completed the final documentation review and handover preparation using the verified existing outputs. Reviewed the accepted results, froze the code without further algorithm changes, updated the final report and daily log, and organised the source files, configuration CSVs, plots, terminal outputs and README.	Final reusable workflow and handover package prepared. The project demonstrates operating-range feasibility in verified windows and clearly records the remaining rigid-mounting, height, vibration and environmental validation requirements.

10. Final Operating Procedure

10.1 Field test

1. Turn on the phone hotspot.
2. Power the i.MX7, PX4Flow and Cube/Pixhawk setup.
3. Wait until the inner i.MX7 LED becomes solid.
4. Keep the setup stationary for 10-15 seconds.
5. Record the complete test on video and perform the planned motion.
6. Do not begin the test if the logger LED does not become solid.

10.2 Download and analysis

7. Connect the laptop to the same phone hotspot.
8. Run the main menu and select the stop-and-download option.
9. Confirm `LOGGER_STOPPED` and that the inner LED turns OFF.
10. Verify that the new CSV is present in a folder created specifically for OF logs
11. Add the corresponding Cube BIN and video timing details.
12. Run window validation and generate the required plots.

11. Future Plans and Recommended Next Steps

The present work establishes a functional PX4Flow logging, synchronization and validation framework. The following activities are recommended before considering flight-level or production integration on the Krishak 2i platform.

11.1 Rigid vehicle integration

Mount the PX4Flow securely on the drone with a fixed downward-facing orientation. The mounting should minimize vibration and prevent changes in sensor angle or height during operation.

During Cube-side integration, configure the measured sensor position and orientation using:

FLOW_POS_X = +0.005 m

FLOW_POS_Y = +0.015 m

FLOW_POS_Z = +0.080 m

FLOW_ORIENT_YAW = verified mounting orientation

The exact orientation and axis signs should be confirmed using controlled forward, rightward, pitch, roll and yaw movements.

11.2 Continuous height measurement

Replace the fixed height used in the current post-processing calculation with a live downward range measurement.

A compatible rangefinder should provide continuous camera-to-ground distance so that optical-flow angular rate can be converted into velocity using the actual height at each sample. This is expected to reduce errors caused by height variation during motion.

11.3 Direct Cube/Pixhawk integration

Connect the PX4Flow directly to the Cube through the supported I²C interface and verify the ArduPilot optical-flow backend.

The integration should confirm:

- Correct PX4Flow detection and update rate
- Correct optical-flow axis orientation
- Valid quality values
- Correct gyro compensation
- Valid rangefinder input
- Optical-flow data availability in Cube BIN logs
- Proper EKF fusion and innovation behaviour

The independent i.MX7 logger may continue to operate in parallel as a raw-data diagnostic and validation source.

11.4 Controlled operating-range validation

Conduct repeatable tests across the Krishak 2i operating-speed range of approximately 4–5 m/s.

Future tests should use:

- Rigid mounting
- Constant and measured height
- A straight and measured test path
- Video-confirmed motion intervals
- A calibrated speed reference, wheel encoder or other independent ground truth
- Multiple repeated runs at each speed

Nominal commanded or speedometer values should be identified as operating-point references unless independently verified.

11.5 Surface, lighting and environmental testing

Optical-flow performance should be evaluated over representative agricultural surfaces, including:

- Soil
- Grass
- Crop canopy
- Wet and dry ground
- Low-texture surfaces
- High-contrast surfaces

Tests should also cover different lighting conditions, shadows, direct sunlight, dust, vibration and pesticide-spray exposure.

11.6 Rotational and geometric compensation validation

The implemented geometric lever-arm correction should be retained as an optional feature and evaluated using rigidly mounted rotational tests.

Pitch, roll and yaw should be tested separately while the vehicle remains translationally stationary. Corrected and uncorrected outputs should be compared using:

- Median false velocity
- Mean false velocity
- Maximum or 95th-percentile false velocity
- Sample count
- Optical-flow quality
- EKF innovation values

Yaw-related error should not be represented using a single constant subtraction because it varies with yaw rate, angular acceleration, tracking quality and operating conditions.

11.7 Optical-tracking dropout detection

The processing and flight-control integration should detect periods where optical-flow output collapses or becomes unreliable.

Future logic may use:

- PX4Flow quality
- Frame-count continuity
- Integration timespan
- Sudden disagreement with IMU or EKF estimates
- Minimum and maximum valid flow-rate limits

Invalid measurements should be flagged or excluded from EKF fusion while raw measurements remain preserved for analysis.

11.8 Scale calibration

After rigid mounting and height validation, calibrate the optical-flow X- and Y-axis scale factors using controlled directional movement.

The following ArduPilot parameters may then be evaluated:

FLOW_FXSCALER

FLOW_FYSCALER

Scale calibration should be performed independently for the forward and lateral axes rather than using only total horizontal-speed magnitude.

11.9 EKF tuning and GPS-denied testing

EKF optical-flow parameters should be tuned only after mounting, orientation, scale and rangefinder inputs have been validated.

The recommended progression is:

1. Stationary ground test
2. Controlled translation test
3. Low-altitude hover with GPS available
4. Compare GPS, EKF and optical-flow velocity
5. Temporarily reduce GPS dependence under controlled conditions
6. Test AltHold or Loiter behaviour
7. Proceed to Guided No-GPS testing in SITL
8. Conduct supervised real-flight GPS-denied testing with safety limits

11.10 Sensor suitability assessment

PX4Flow should be treated as a feasibility sensor rather than automatically selected for production integration.

A future comparison may evaluate newer optical-flow sensors based on:

- Maximum usable height
- Maximum measurable velocity
- Low-light performance
- Vibration tolerance
- Surface sensitivity
- Interface compatibility
- Availability and cost
- ArduPilot support
- Environmental protection

12. Final Conclusion

The project progressed from basic sensor communication to an autonomous optical-flow logging and validation system. Controlled measured-distance tests produced approximately 8-9% error over the low-speed calibration range. Four verified nominal 15-20 km/h bike windows met the 20% feasibility criterion with an average absolute error of approximately 11.0%, demonstrating useful operating-range potential under selected clean conditions. Long-duration disturbed tests showed reduced repeatability, particularly during roll and yaw, so rigid mounting and further flight-environment validation are still required.

The most important deliverable is the repeatable workflow: independent logging, synchronized timestamps, LED status, safe passwordless download, explicit time alignment, automatic and manual window processing, and native-rate OF/GPS/EKF analysis. This provides a practical handover foundation for continued GPS-denied velocity development. The project therefore concludes positively at a feasibility and workflow-readiness stage, not as final Krishak 2i integration approval.

13. Reference Links

All Drivers, Post processing links :- [px4flow_refactored](#)

Logs:- GPS: [GPS_logs](#)

OF: [OF_Logs](#)